Predicting Mechanical Properties of Aluminum Alloys

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Introduction

Strategic materials selection for engineering is critical for ensuring that physical components can withstand mechanical stresses and provide the required properties for a given application. For example, using a material that is too brittle in a load-bearing situation may lead to premature failure and breakage. The development of metal alloys for engineering applications has historically been iterative and time-consuming through trial-and-error experimentation of different alloy compositions and heat treatments to meet required properties. Machine learning modeling and simulation are increasingly used in materials science to predict how a theoretical material might behave, or what candidate materials might be able to meet certain requirements. With enough historical data from empirical experimentation as well as reported datasheets from manufacturers, models can be developed to help shortcut the research and development that might otherwise take months or years.

The second capstone project as part of the HarvardX Professional Certificate in Data Science is to utilize R and a publicly available dataset to solve an open-ended problem with a model that is more complex than linear or logistic regression. The chosen problem is to use reported elemental composition and processing condition combinations for a variety of unique aluminum-based alloys to help predict what mechanical properties would be when presented with a new hypothetical aluminum alloy. The dataset used for this project comes from Mendeley Data, a free repository for scientific research data. The three mechanical properties of interest are tensile strength, yield strength, and elongation. The dataset is duplicated and transformed into three versions that each focus on one mechanical property to develop a model. The three tables and models independently predict each property using only the composition and processing data without taking into account the other mechanical properties. Each of the three datasets is split 80/20 into training/test sets and different models are compared to maximize coefficient of determination (R^2, the square of correlation coefficient r) when applied to the actual ratings in the final holdout test set.

This report outlines the data cleaning, exploration, and visualizations used to gain insight into variables that could be used to develop a machine learning regression model. The resulting models are applied to alloys with missing data to help predict what the expected properties would be before discussing limitations and future work.

Methods and Analysis

A publicly available .csv file from Mendeley Data [1] is downloaded for the purposes of this project. The raw dataset contains 31 columns relating to 1154 anonymous alloys of varying conditions:

- **Column 1.** “X”: integer index unique to each anonymous alloy.
- **Column 2.** “Processing”: a character string indicating the thermal and/or mechanical manufacturing processes used to make the aluminum alloy to achieve the recorded mechanical property values.
- **Columns 3-27.** 25 columns for each of 25 elements of interest that determine an alloy’s composition. Elements include “Al” (base aluminum) and the following alloying elements: “Ag”, “Al”, “B”, “Be”, “Bi”, “Cd”, “Co”, “Cr”, “Cu”, “Er”, “Eu”, “Fe”, “Ga”, “Li”, “Mg”, “Mn”, “Ni”, “Pb”, “Sc”, “Si”, “Sn”, “Ti”, “V”, “Zn”, and “Zr”.
- **Column 28.** “Elongation....”: a number representing the percentage of a starting length of sample material an alloy can be stretched before the material breaks (total elongation) [2].
- **Column 29.** “Tensile.Strength..MPa.”: a number representing the amount of tension in megapascals (MPa) that a sample material can be stretched before breaking [2].
- **Column 30.** “Yield.Strength..MPa.”: a number representing the amount of tension in MPa that a sample material can be stretched before deformation starts to become permanent [2].
- **Column 31.** “class”: a character (displayed as an integer) that represents the family of aluminum alloys that an anonymous aluminum alloy belongs to.

An example engineering stress-strain curve [3] helps visualize the meaning of the mechanical properties, where the amount of pulling/tensile strain applied to a sample is on the x-axis and the amount of pulling pressure needed to maintain a certain strain when the sample is gradually stretched is on the y-axis.

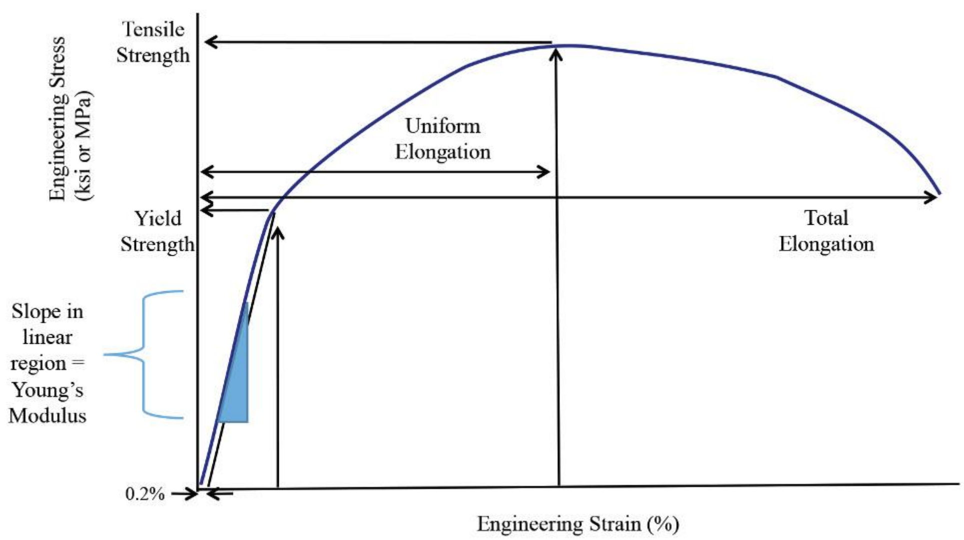


Figure 1: Schierman, Mark (2024), ’Getting to know more about the metal you are forming’, The Fabricator, thefabricator.com

Starting from zero strain, the curve is linear up until the yield strength. Release from tension at this linear section will allow the sample to return to its original shape - this is considered the elastic segment. At the yield strength, the curve stops being linear due to permanent strain - release will go back towards zero with the same slope as the elastic segment but will be offset by the permanent strain. At the tensile strength, the curve starts going down as this is the point where the sample can no longer withstand any more stress and will start approaching the breaking point. The end of the curve is where the sample breaks, with the x-axis value defining total elongation.

The mechanical properties of alloys are primarily modulated by both elemental composition and the physical manufacturing processes. Pure aluminum is soft and may not have the strength to meet the requirements of a given engineering application [4]. Other elements are purposely added to pure aluminum to create stronger alloys that cannot otherwise be achieved. Alloy designations are often 4-digit numbers where the first digit is the series number representing families of alloys that share dominant alloying elements [5]. In the provided dataset this first digit is considered the “class”. These families are generally explained in this table:

Aluminum Alloy Group	Principal Alloying Element
1000 Series	Pure Aluminum - mininum 99.0+% purity
2000 Series	Copper
3000 Series	Manganese
4000 Series	Silicon
5000 Series	Magnesium
6000 Series	Magnesium and Silicon
7000 Series	Zinc
8000 Series	Other Elements (silicon, iron, copper, magnesium, zinc, and boron,...)

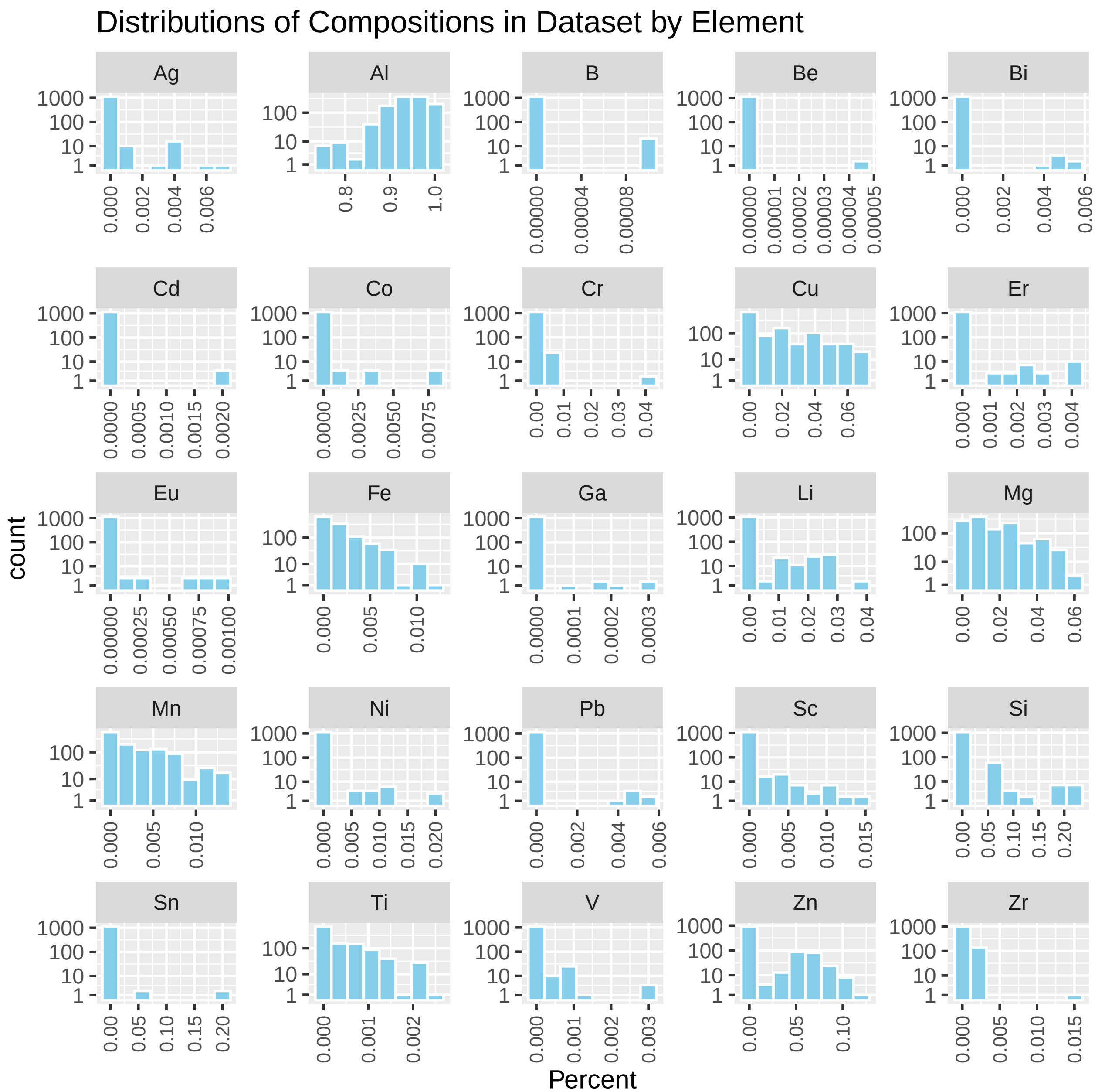
Figure 2: The Aluminum Association (2024), “’Aluminum Alloy Series Designations’, GlobalSpec, https://insights.globalspec.com/images/assets/319/5319/AluminumAssociationAluminumAlloySeriesDesignations.png

In the provided dataset, the “class” column represents the series number of each alloy. The following table summarizes the incidence of each series in the dataset:

class	n
1	145
2	372
3	111
4	103
5	197
6	86
7	62
8	75
outlier	3

Each series has at least 62 alloys represented in the dataset. Three alloys are noted as “outlier”, though 8xxx series is defined as “Other” alloys that do not fit in other alloy series. In order to not overfit models with an “outlier” classification with such few entries, it is best and convenient to consider outliers as series 8 closest to industry definitions.

A series of histograms of each element covered by all of the provided alloys helps visualize the relative amounts of alloying elements used in the wide variety of compositions. Aluminum (Al) content is typically between 75-99.9%. The contents of some alloying elements are on the order of 1-40% (like Cu, Mg, Si) while others are on the order of parts per million (like B, Be, Co). The effects of some alloying elements may have disproportionately strong effects on properties.



Properties are modulated by manufacturing processes, also known as tempers [4]. Heat treatments at different temperatures and durations are commonly used to influence how the elements are arranged at microscopic levels. Mechanical work can also influence properties by strategically forcing the microscopic structure into arrangements that are stronger than before. The following table summarizes the processing descriptions and incidence in the dataset:

Processing	n
Artificial aged	21
Naturally aged	18
No Processing	141
Solutionised	4
Solutionised + Artificially peak aged	375
Solutionised + Artificially over aged	214
Solutionised + Cold Worked + Naturally aged	57
Solutionised + Naturally aged	86
Strain Hardened (Hard)	197
Strain hardened	41

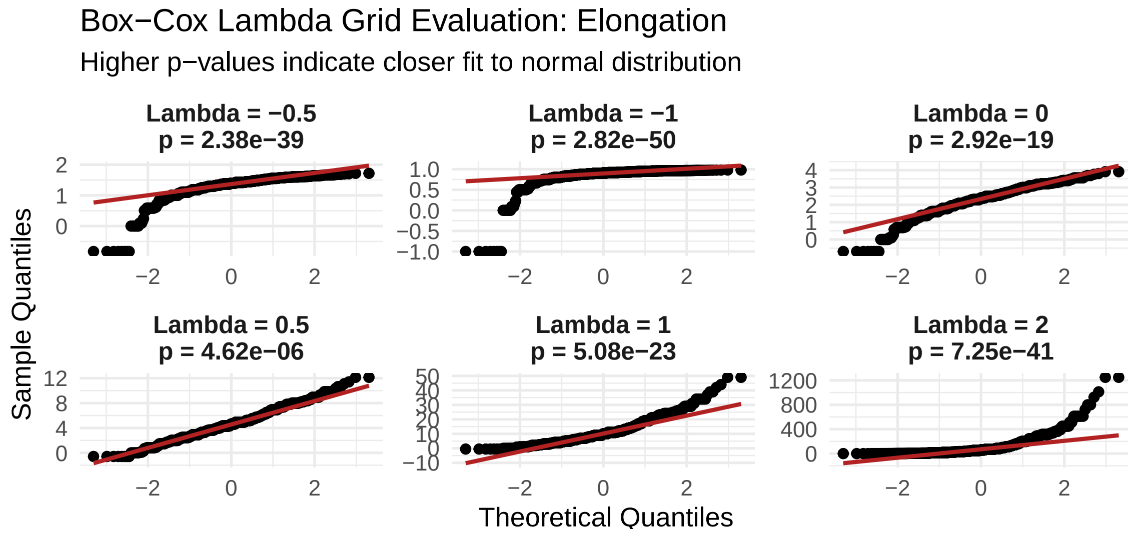
As provided, the processing descriptions can consist of a single processing step or multiple steps. There are also some misspellings that should be corrected. One-hot encoding can help break down the combinations in the "Processing" column into a few new and simplified boolean columns indicating whether a type of processing is present to use as predictors.

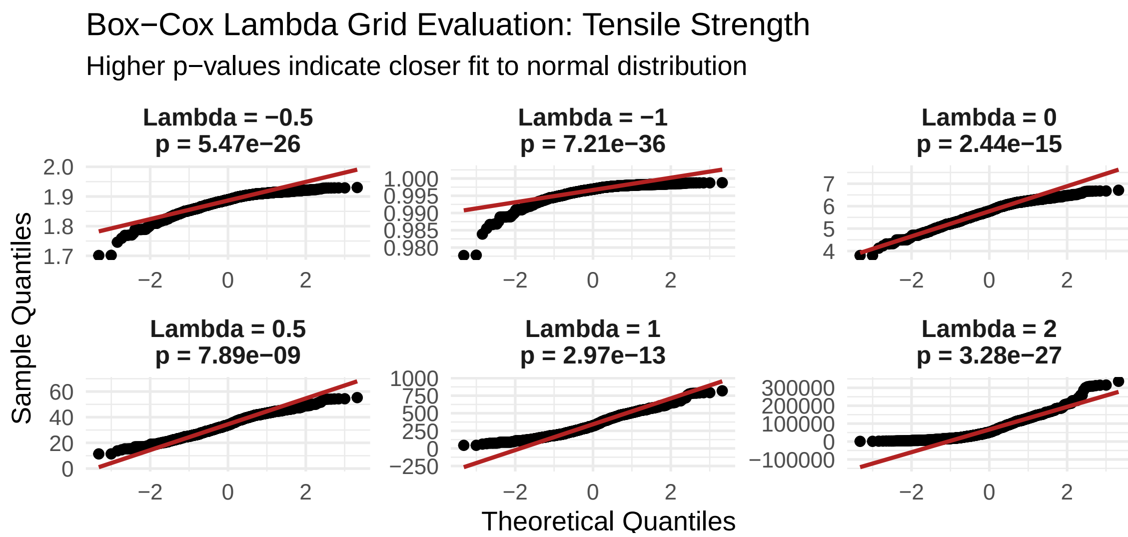
Prior to building any models, it is also important to inspect the normality of the mechanical property data and see if any data transformations would be useful for constructing better models than just using the values as-provided. For example, perhaps distribution of the provided elongation values becomes closer to a normal distribution after taking the natural logarithm of each value. A Box-Cox transformation analysis can help decide what kind of transformation would be appropriate for modeling the data [6]. The following formula is applied to each value of the mechanical properties, screening through several different lambdas to find an optimal transformation:

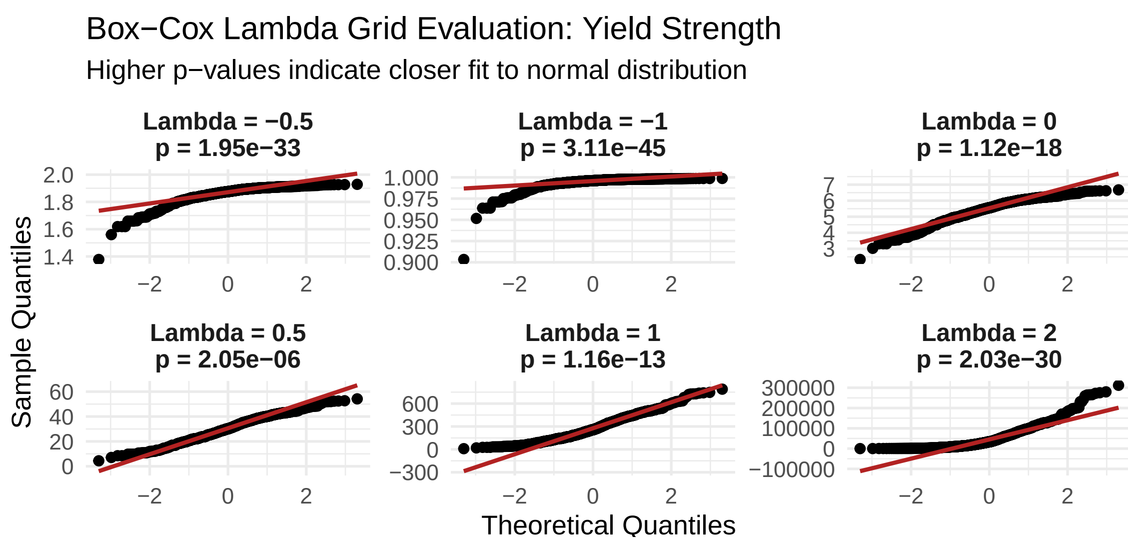
$$y^{(\lambda)} = \begin{cases} \frac{y^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(y) & \text{if } \lambda = 0 \end{cases}$$

For simplicity only six lambda values were screened: a lambda of -1 results in an inverse transform, -0.5 results in an inverse square root transform, 0 results in a natural logarithm transform, 0.5 results in a square root transform, 1 results in no transformation (data is normally distributed as-is), and 2 results in a square transform.

The following quantile plots show how close to normal the population of each property is after transformation, with p-values indicating statistical significance of the null hypothesis that the distribution is normal:







Although p-values are less than 0.05 for all plots made, the use of lambda = 0.5 consistently yielded distributions that were more normally distributed than the original data. This implies that a square root transform would be appropriate to build models upon, with model predictions ultimately being squared to get final predictions. This conveniently helps eliminate the potential of negative values of strengths and elongations being predicted, as they are physically impossible.

The variables provided in the raw dataset can either be directly used or cleaned into forms of the same information that is more compatible for building models. For the purposes of exploration, the raw dataset is modified into a cleaner format. Modifications include:

1. Renaming "Elongation...." to simply "elongation".
2. Renaming "Tensile.Strength..MPa." to simply "tensile_strength".
3. Renaming "Yield.Strength..MPa." to simply "yield_strength".
4. Renaming "class" to "series" to prevent confusion with R syntax and be more aligned with aluminum industry nomenclature.
5. One-hot encode the 10 unique combinations of processing steps into four binary True or False columns that show if solutionisation is part of the "Processing" string, if natural ageing is part of the "Processing" string, if artificial ageing is part of the "Processing" string, and if strain hardening is part of the "Processing" string. By default, an alloy noted as "No Processing" will show as False for each of the four types of processes.
6. Changing "outlier" values in the "series" column to series 8 in accordance with the aluminum industry referring to series 8 as a miscellaneous category.
7. Making a new "balance" column subtracting all elemental fractions from 1 to get a fraction of the alloys unaccounted for with the 25 elements used in the dataset.
8. Adding square root transformed "sqrt_elongation", "sqrt_tensile_strength", and "sqrt_yield_strength" columns alongside the original property columns, based off of Box-Cox observations.

The general approach to this project is to compare several machine learning regression algorithms to see which one or which ensemble combination maximizes the coefficient of determination (R^2, the square of correlation coefficient r). R^2 is calculated by fitting a line to predicted vs. actual property values to get a correlation coefficient r. The square of r becomes a metric to maximize that represents the proportion of variance in each mechanical property that is explained by the prediction variables in the regression models.

These metrics are used to evaluate three separate models: one for tensile strength, one for yield strength, and one for elongation. Many alloys are missing results for one or two of the three mechanical properties, so to build the models the raw table is filtered into three tables:

1. Tensile strength table containing only the series, one-hot process conditions, elements, and tensile strengths for alloys that have tensile strengths reported.
2. Yield strength table containing only the series, one-hot process conditions, elements, and yield strengths for alloys that have yield strengths reported.
3. Elongation table containing only the series, one-hot process conditions, elements, and elongations for alloys that have elongations reported.

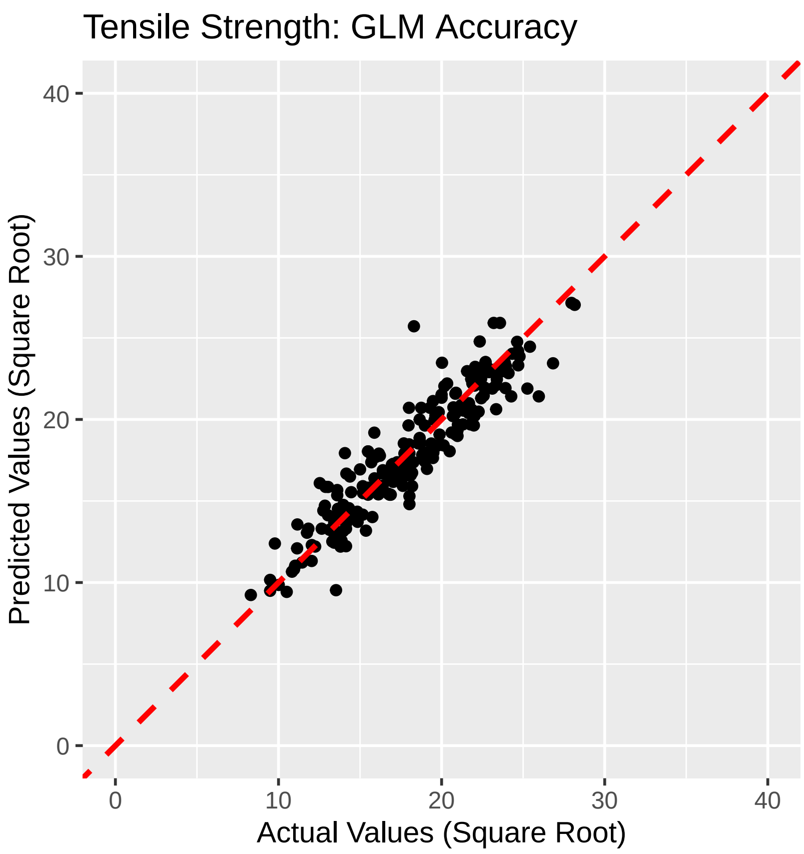
After splitting into three tables and filtering out rows with empty values in their respective property columns, the tensile strength table is left with 1121 rows, the yield strength table is left with 1045 rows, and the elongation table is left with 1046 rows. Given that each dataset has roughly 1000 entries, an 80% training and 20% testing partition of each small-medium dataset is implemented for model cross-validation with R CARET package machine learning models, as the train() function from the package already incorporates training/validation partitioning of the training set used as the input. This ratio provides at least 200 entries to do final testing while having sufficient data in the training set to accommodate further partitioning. The partitioned datasets are then ready to use across a variety of modelling algorithms to determine which model or ensemble of models is best for each mechanical property. For the remainder of this report, the raw code is hidden to reduce clutter - please refer to an accompanying .R file for the source code for the models.

Results - Tensile Strength

The approach to cross-validation of different models for predicting tensile strength is to:

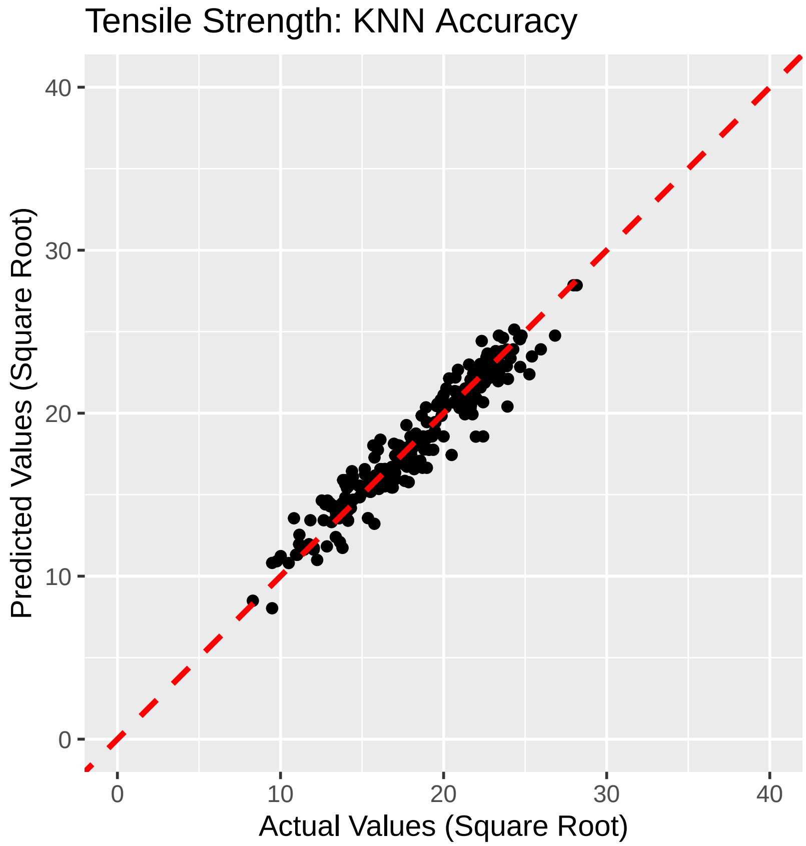
1. Train each algorithm on its own using the square root transformed tensile strengths - the methods tested include generalized linear modeling (glm), K-nearest neighbors (knn), regression trees (rpart), random forest (rf), extreme gradient boosted trees (xgbTree), and neural network (nnet). These are common regression methods that cover a broad variety of machine learning approaches to try out and see which are most effective for predicting properties of aluminum alloys.
2. Train all of them together as an ensemble and find the relative importance of each method.
3. Downselect to ensembles with the most promising algorithms.
4. Compare the R^2 for all of the above and determine which model has the best performance on square root transformed tensile strengths.
5. Square the predictions of the final model to be in units that actually represent the predicted tensile strengths of the alloys.

The first model is to simply use a generalized linear model as a baseline. While this is not expected to perform as well as more advanced machine learning methods, the model provides a modest training R^2 of 0.626. When the model is applied to the test set, R^2 is a respectable 0.866.



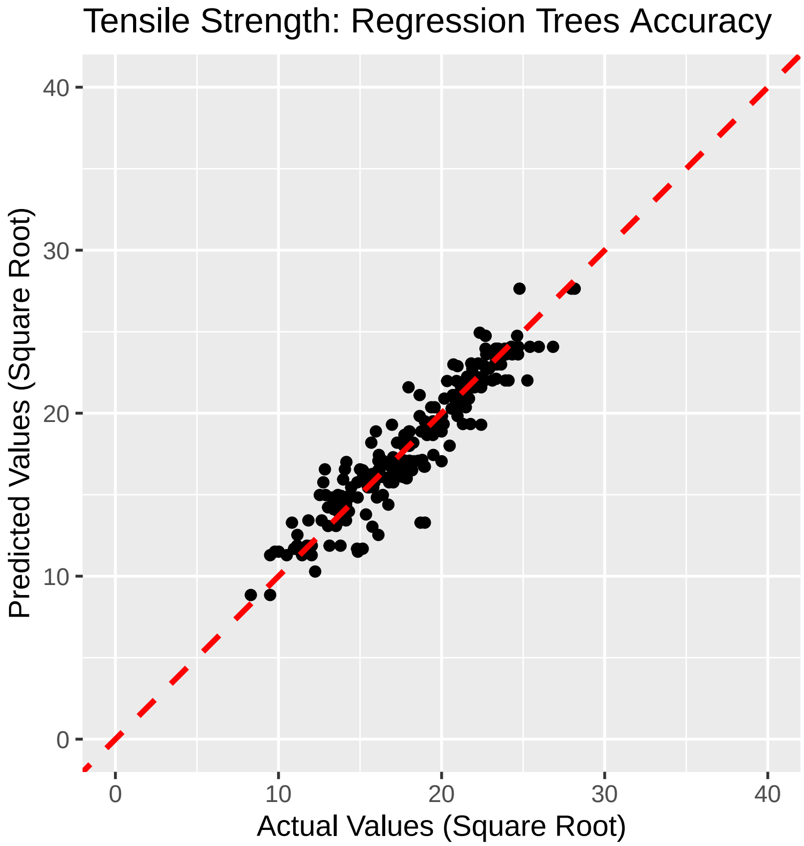
TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866

The next model is to try K-nearest neighbors. After tuning the model provides an improved training R^2 of 0.894 and a testing R^2 of 0.924.



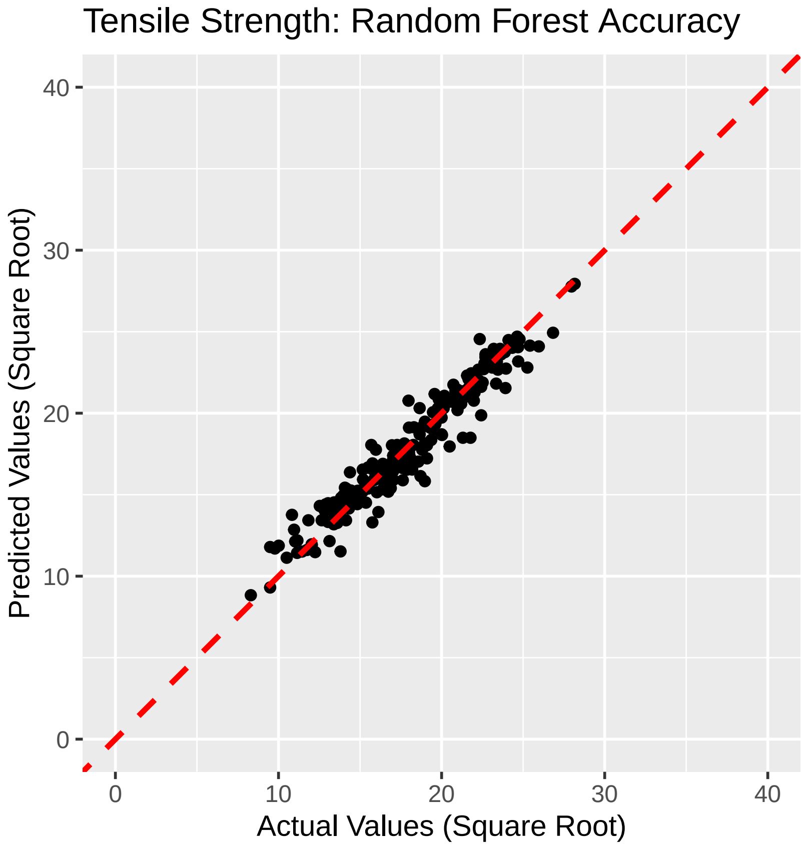
TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866
KNN	0.894	0.924

The next model is to try regression trees. After tuning the model provides a slightly worse training R^2 of 0.877 and a testing R^2 of 0.880.



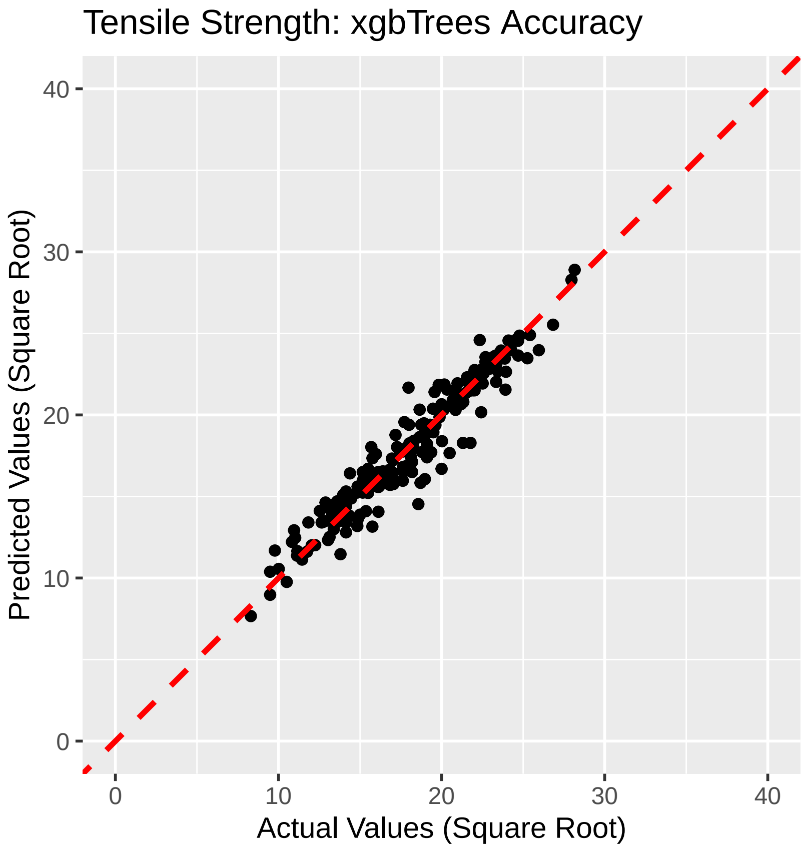
TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866
KNN	0.894	0.924
Regression Trees	0.877	0.880

The next model is to try random forest. After tuning the model performs very well providing a training R^2 of 0.924 and a testing R^2 of 0.934.



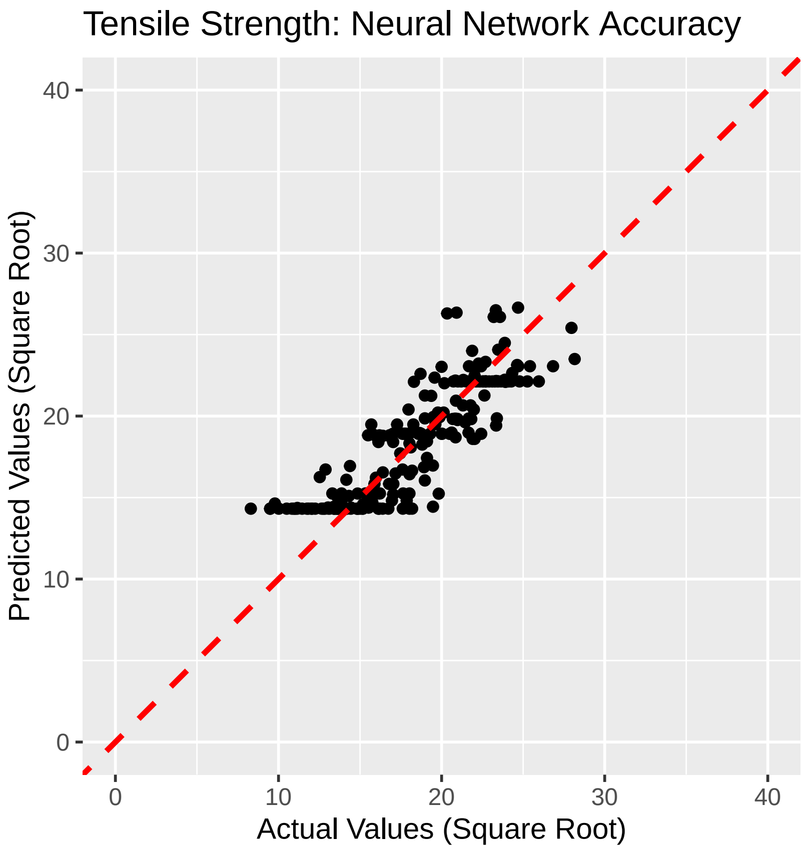
TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866
KNN	0.894	0.924
Regression Trees	0.877	0.880
Random Forest	0.924	0.934

The next model is to try extreme gradient boosted trees. After tuning the model also has excellent results with a training R^2 of 0.923 and a testing R^2 of 0.928.



TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866
KNN	0.894	0.924
Regression Trees	0.877	0.880
Random Forest	0.924	0.934
eXtreme Gradient Boosted Trees	0.923	0.928

The next model is to try neural networks. After tuning the model provides a poor training R^2 of 0.845 and a testing R^2 of 0.739. The predictions tend to cluster around a few values but are not differentiated or granular enough between those values to provide meaningful predictions.

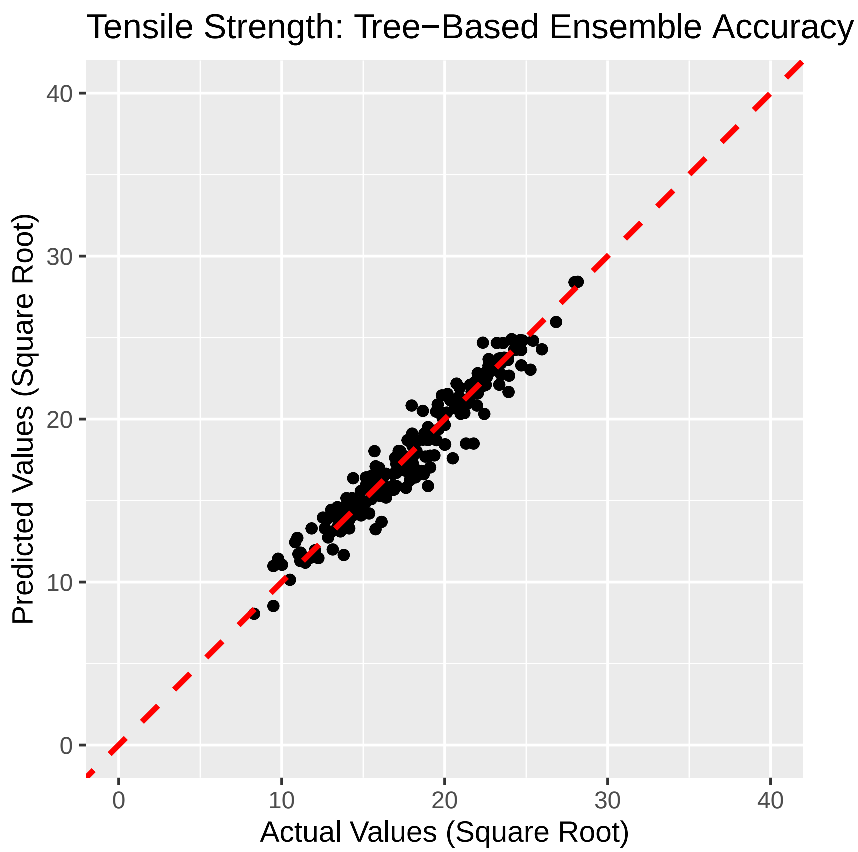


TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866
KNN	0.894	0.924
Regression Trees	0.877	0.880
Random Forest	0.924	0.934
eXtreme Gradient Boosted Trees	0.923	0.928
Neural Network	0.845	0.739

Some models on their own perform better than others, but combining some or all together into an ensemble model could produce better predictions than any one method alone. To evaluate the relative strengths of each method for predicting tensile strength, a generalized linear meta-model was applied to all six models and the resulting model importance summary was used to downselect the random forest and extreme gradient boosted trees tree-based models as the most dominant models that could be used for a final ensemble model.

```
## The following models were ensembled: glm, knn, rpart, rf, xgbTree, nnet
##
## Model Importance:
##     glm     knn   rpart      rf xgbTree    nnet
##  0.0029  0.0580  0.0258  0.3453  0.5681  0.0000
##
## Model accuracy:
##    model_name metric     value        sd
##        <char> <char>     <num>     <num>
## 1:   ensemble   RMSE  1.008183 0.1519305
## 2:        glm   RMSE  1.551949 0.2653417
## 3:        knn   RMSE  1.372995 0.2195270
## 4:      rpart   RMSE  2.767299 0.2739531
## 5:         rf   RMSE  1.057119 0.1220156
## 6:    xgbTree   RMSE  1.055591 0.1355030
## 7:       nnet   RMSE 17.593648 0.1258358
```

A downselected ensemble model provides the best training R^2 of 0.942 and a testing R^2 of 0.940.



TensileStrengthModel	TrainRsquared	TestRsquared
GLM	0.626	0.866
KNN	0.894	0.924
Regression Trees	0.877	0.880
Random Forest	0.924	0.934
eXtreme Gradient Boosted Trees	0.923	0.928
Neural Network	0.845	0.739
Tree-Based Ensemble	0.942	0.940

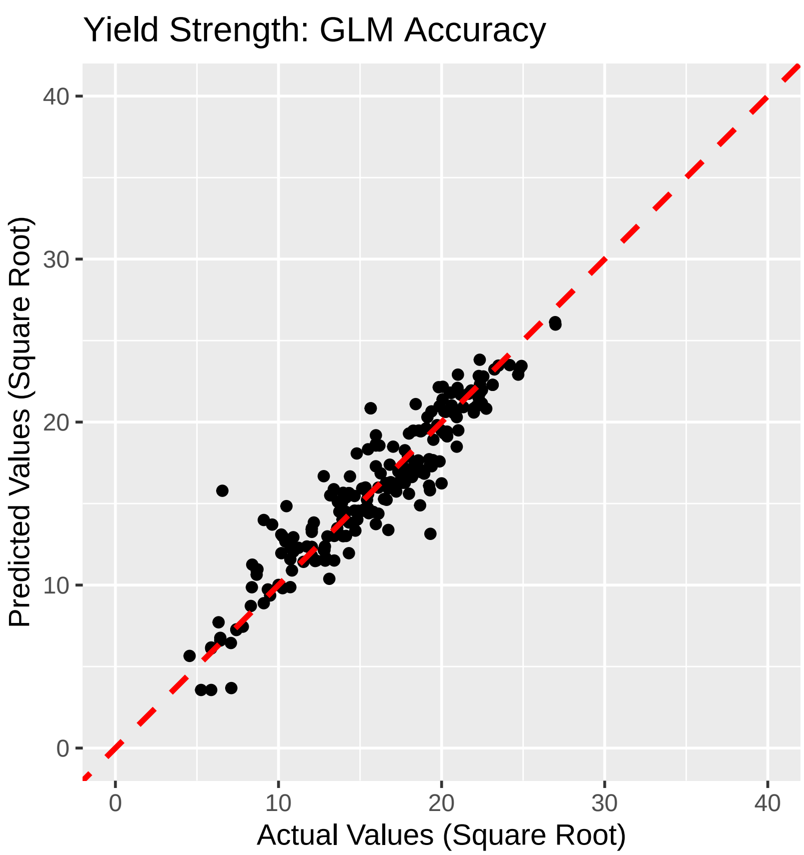
The tree-based ensemble is deemed the most appropriate model for predicting tensile strengths due to its excellent performance in both training and testing. Random forest on its own is very close behind as an alternative non-ensemble model.

Results - Yield Strength

As was done with tensile strength, the approach to cross-validation of different models for predicting yield strength is to:

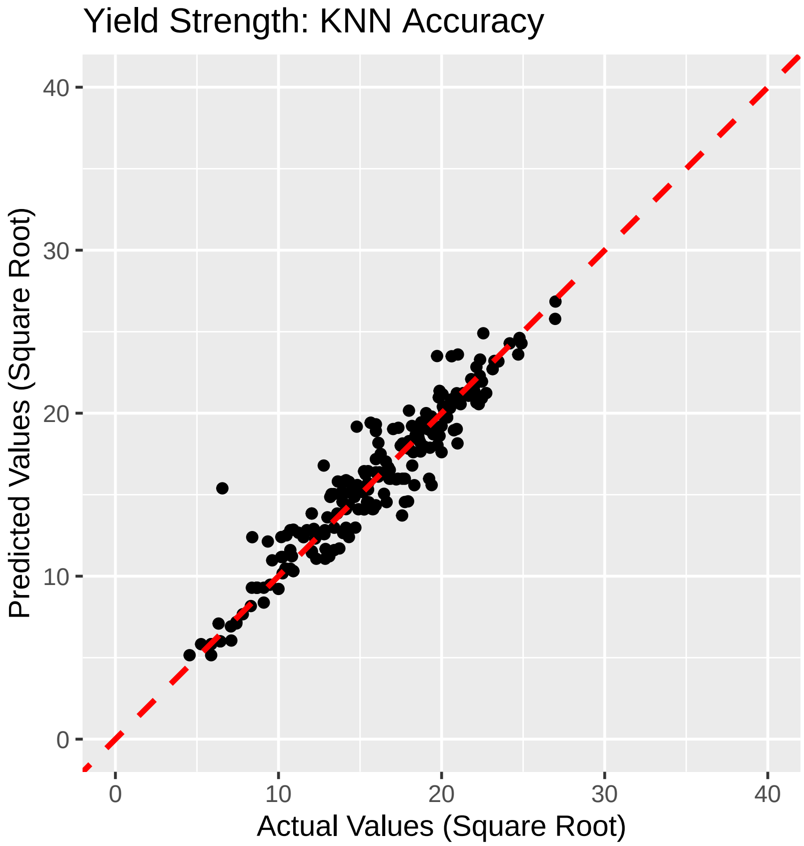
1. Train each algorithm on its own using the square root transformed yield strengths - the methods tested include generalized linear modeling (glm), K-nearest neighbors (knn), regression trees (rpart), random forest (rf), extreme gradient boosted trees (xgbTree), and neural network (nnet). These are common regression methods that cover a broad variety of machine learning approaches to try out and see which are most effective for predicting properties of aluminum alloys.
2. Train all of them together as an ensemble and find the relative importance of each method.
3. Downselect to ensembles with the most promising algorithms.
4. Compare the R^2 for all of the above and determine which model has the best performance on square root transformed yield strengths.
5. Square the predictions of the final model to be in units that actually represent the predicted yield strengths of the alloys.

The first model is to simply use a generalized linear model as a baseline. While this is not expected to perform as well as more advanced machine learning methods, the model provides a modest training R^2 of 0.658. When the model is applied to the test set, R^2 is a respectable 0.864.



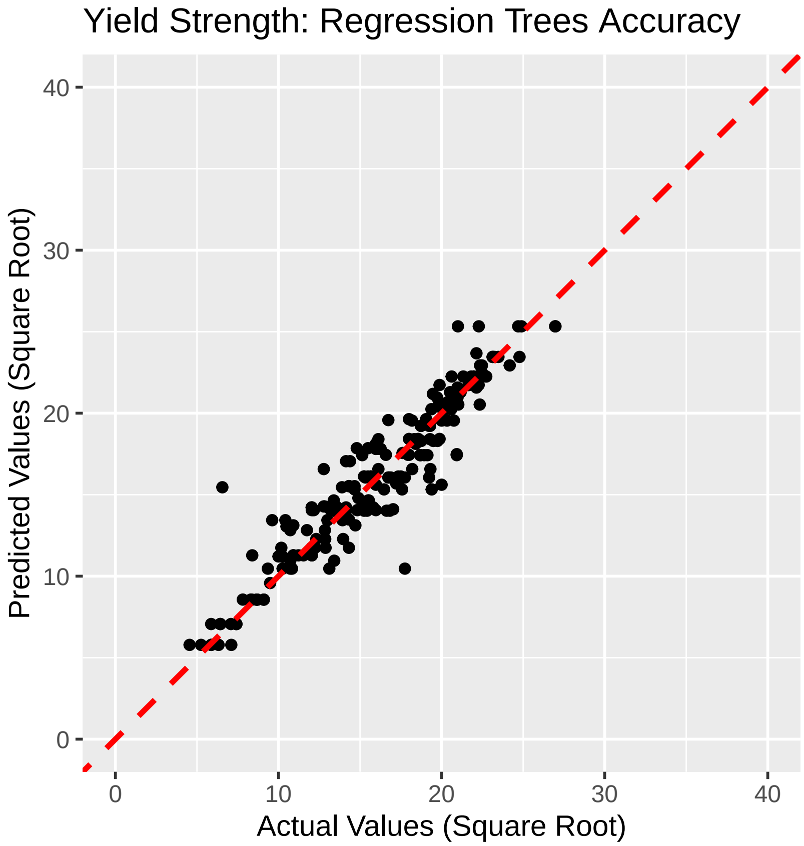
YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864

The next model is to try K-nearest neighbors. After tuning the model provides an improved training R^2 of 0.860 and a testing R^2 of 0.899.



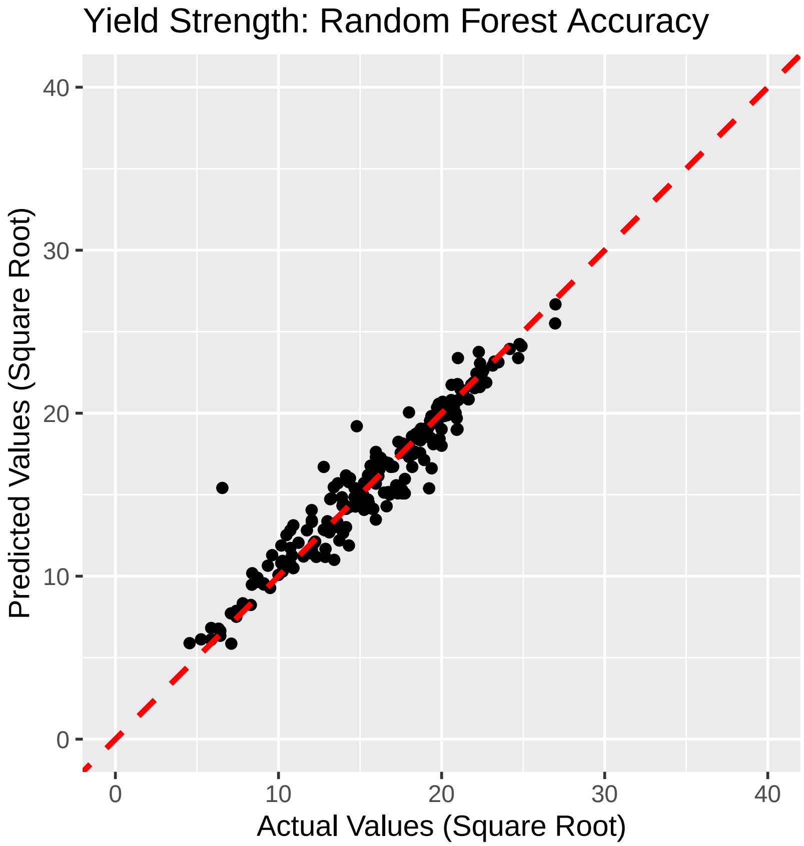
YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864
KNN	0.860	0.899

The next model is to try regression trees. After tuning the model provides a slightly worse training R^2 of 0.821 and a testing R^2 of 0.886.



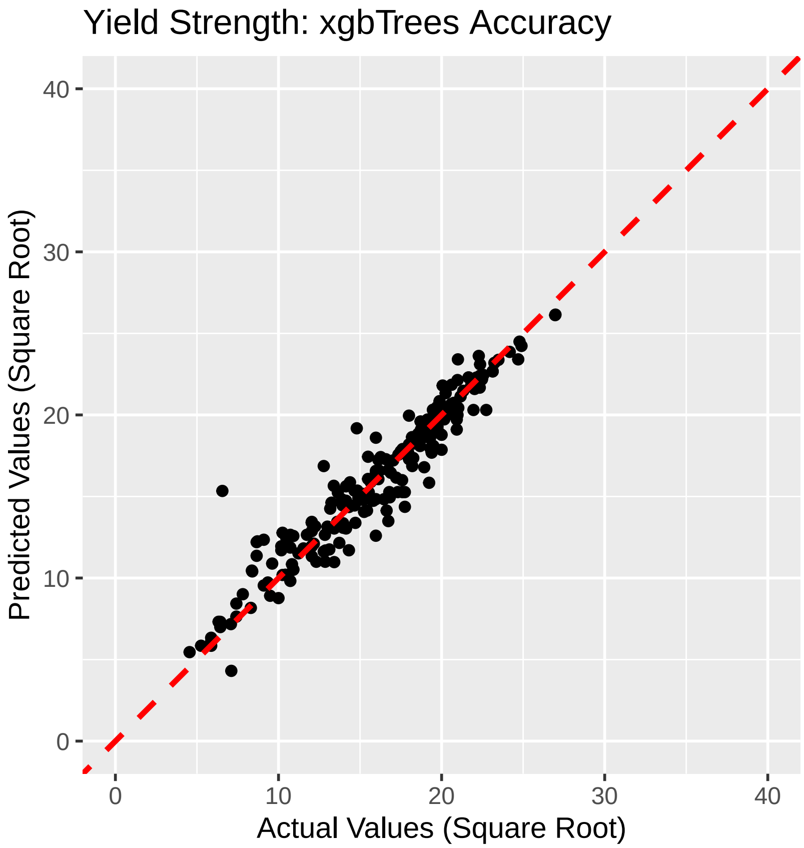
YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864
KNN	0.860	0.899
Regression Trees	0.821	0.886

The next model is to try random forest. After tuning the model performs very well providing a training R^2 of 0.893 and a testing R^2 of 0.926.



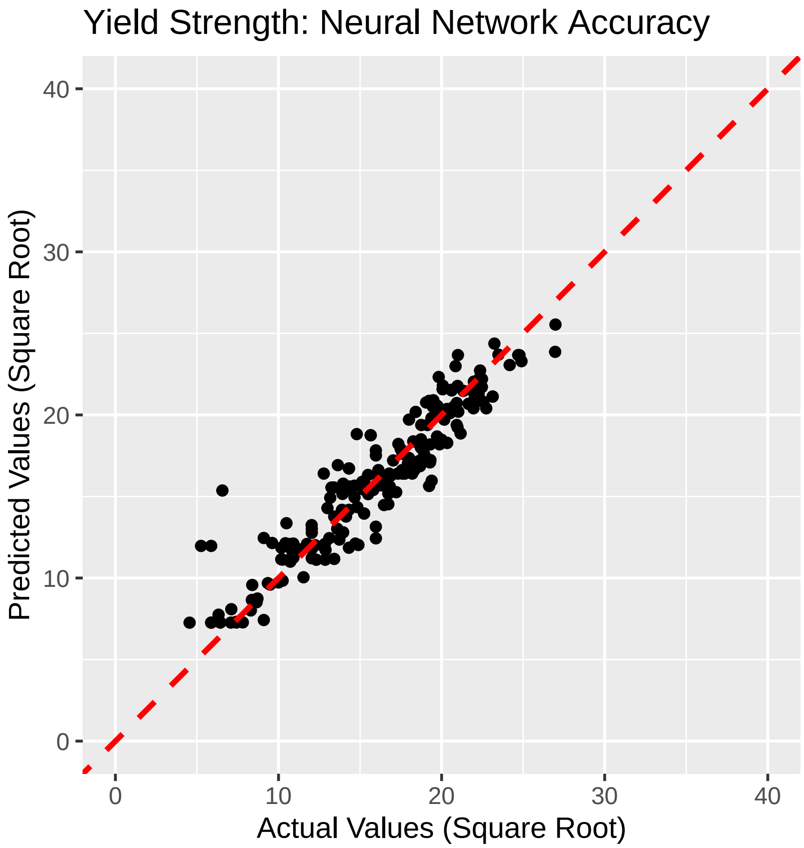
YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864
KNN	0.860	0.899
Regression Trees	0.821	0.886
Random Forest	0.893	0.926

The next model is to try extreme gradient boosted trees. After tuning the model also has excellent results with a training R^2 of 0.891 and a testing R^2 of 0.911.



YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864
KNN	0.860	0.899
Regression Trees	0.821	0.886
Random Forest	0.893	0.926
eXtreme Gradient Boosted Trees	0.891	0.911

The next model is to try neural networks. After tuning the model provides a training R^2 of 0.826 and a testing R^2 of 0.880.

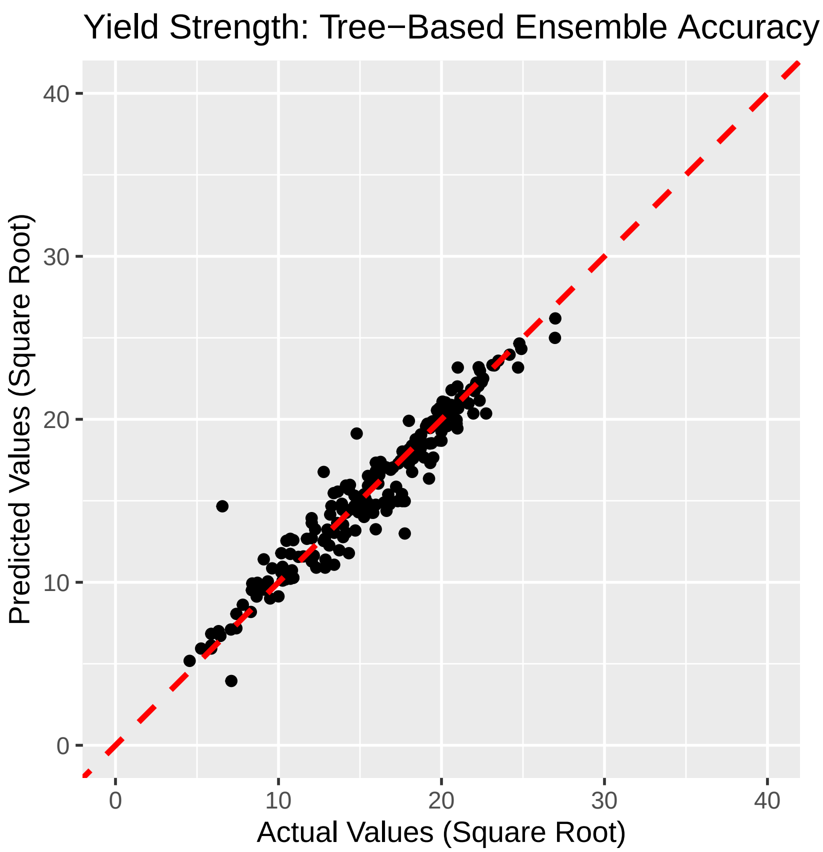


YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864
KNN	0.860	0.899
Regression Trees	0.821	0.886
Random Forest	0.893	0.926
eXtreme Gradient Boosted Trees	0.891	0.911
Neural Network	0.826	0.880

Some models on their own perform better than others, but combining some or all together into an ensemble model could produce better predictions than any one method alone. To evaluate the relative strengths of each method for predicting yield strength, a generalized linear meta-model was applied to all six models and the resulting model importance summary was used to downselect the random forest and extreme gradient boosted trees tree-based models as the most dominant models that could be used for a final ensemble model.

```
## The following models were ensembled: glm, knn, rpart, rf, xgbTree, nnet
##
## Model Importance:
##     glm     knn   rpart      rf xgbTree    nnet
##  0.0141  0.0093  0.0210  0.4610  0.4946  0.0000
##
## Model accuracy:
##    model_name metric     value        sd
##        <char> <char>     <num>     <num>
## 1:   ensemble   RMSE  1.272893 0.1091594
## 2:        glm   RMSE  1.863619 0.2857619
## 3:        knn   RMSE  1.666467 0.1882664
## 4:      rpart   RMSE  3.100842 0.2316862
## 5:         rf   RMSE  1.316084 0.1964386
## 6:    xgbTree   RMSE  1.356047 0.1813001
## 7:       nnet   RMSE 15.849374 0.1431011
```

A downselected ensemble model provides the best training R^2 of 0.925 and a testing R^2 of 0.925.



YieldStrengthModel	TrainRsquared	TestRsquared
GLM	0.658	0.864
KNN	0.860	0.899
Regression Trees	0.821	0.886
Random Forest	0.893	0.926
eXtreme Gradient Boosted Trees	0.891	0.911
Neural Network	0.826	0.880
Tree-Based Ensemble	0.925	0.925

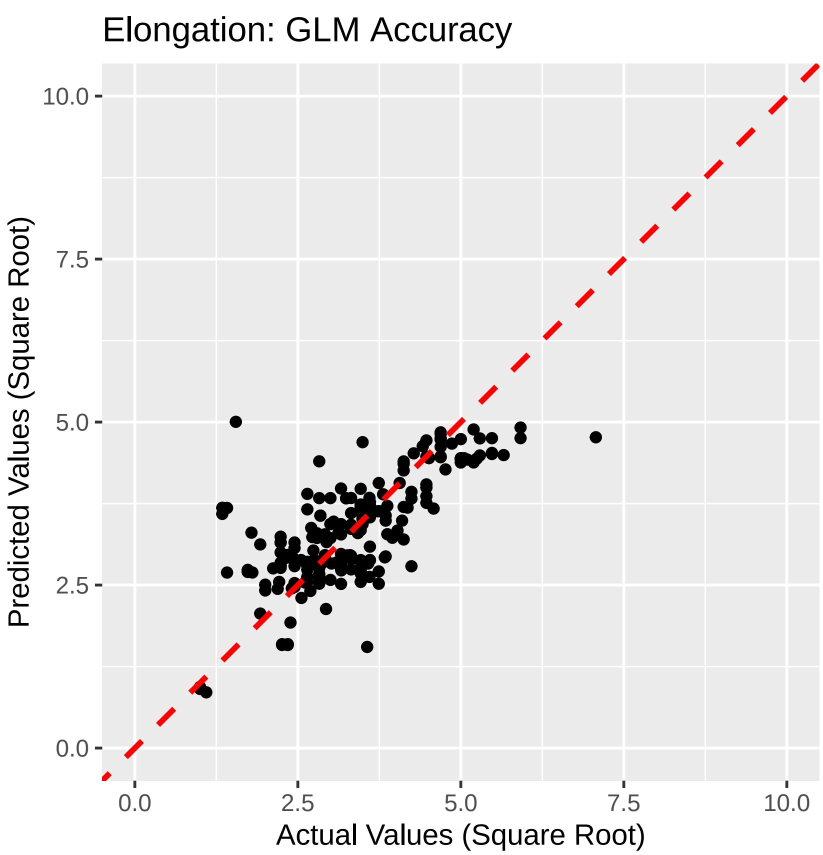
The tree-based ensemble is deemed the most appropriate model for predicting yield strengths due to its excellent performance in both training and testing. Random forest on its own is very close behind as an alternative non-ensemble model.

Results - Elongation

As was done with the strength properties, the approach to cross-validation of different models for predicting elongation is to:

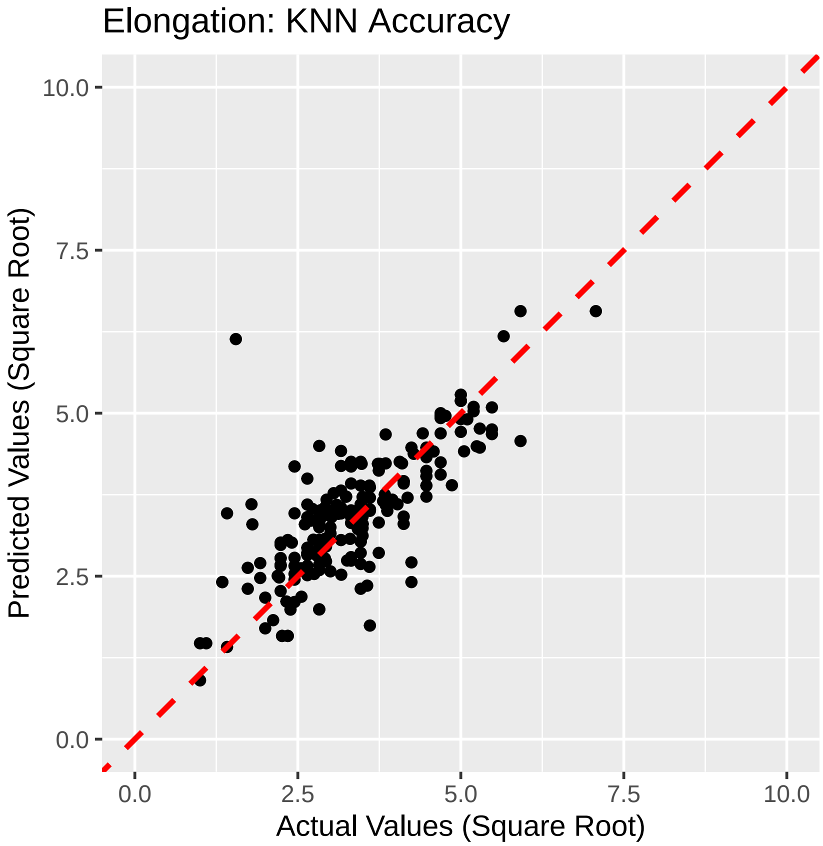
1. Train each algorithm on its own using the square root transformed elongations - the methods tested include generalized linear modeling (glm), K-nearest neighbors (knn), regression trees (rpart), random forest (rf), extreme gradient boosted trees (xgbTree), and neural network (nnet). These are common regression methods that cover a broad variety of machine learning approaches to try out and see which are most effective for predicting properties of aluminum alloys.
2. Train all of them together as an ensemble and find the relative importance of each method.
3. Downselect to ensembles with the most promising algorithms.
4. Compare the R^2 for all of the above and determine which model has the best performance on square root transformed elongations.
5. Square the predictions of the final model to be in units that actually represent the predicted elongations of the alloys.

The first model is to simply use a generalized linear model as a baseline. While this is not expected to perform as well as more advanced machine learning methods, the model provides a poor training R^2 of 0.285. When the model is applied to the test set, R^2 is a modest 0.547.



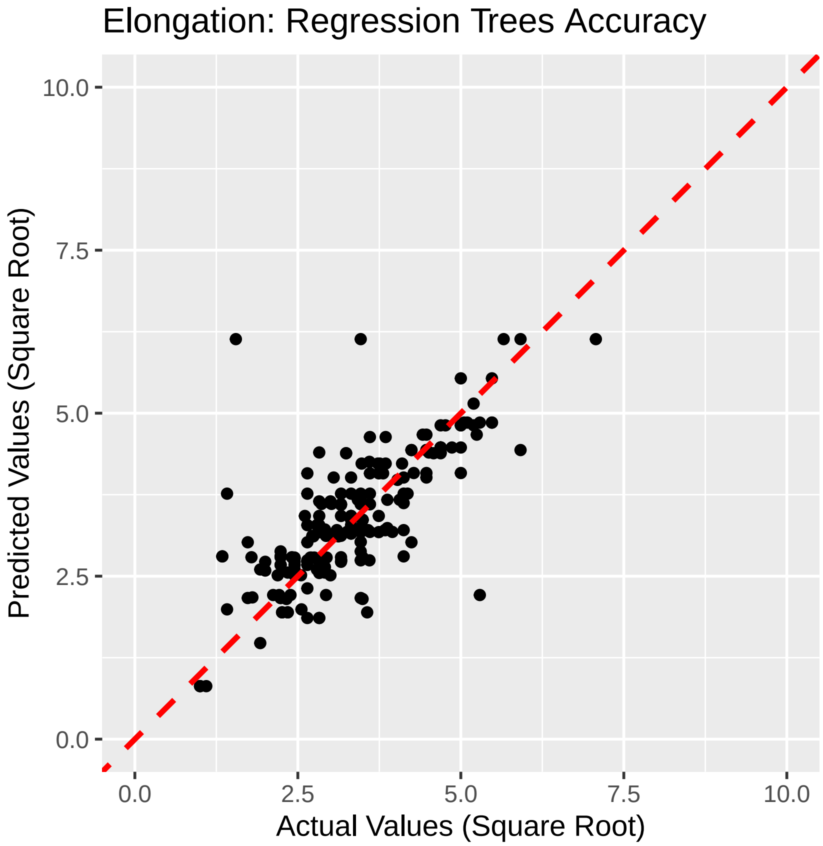
ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547

The next model is to try K-nearest neighbors. After tuning the model provides an improved training R^2 of 0.624 and a testing R^2 of 0.586.



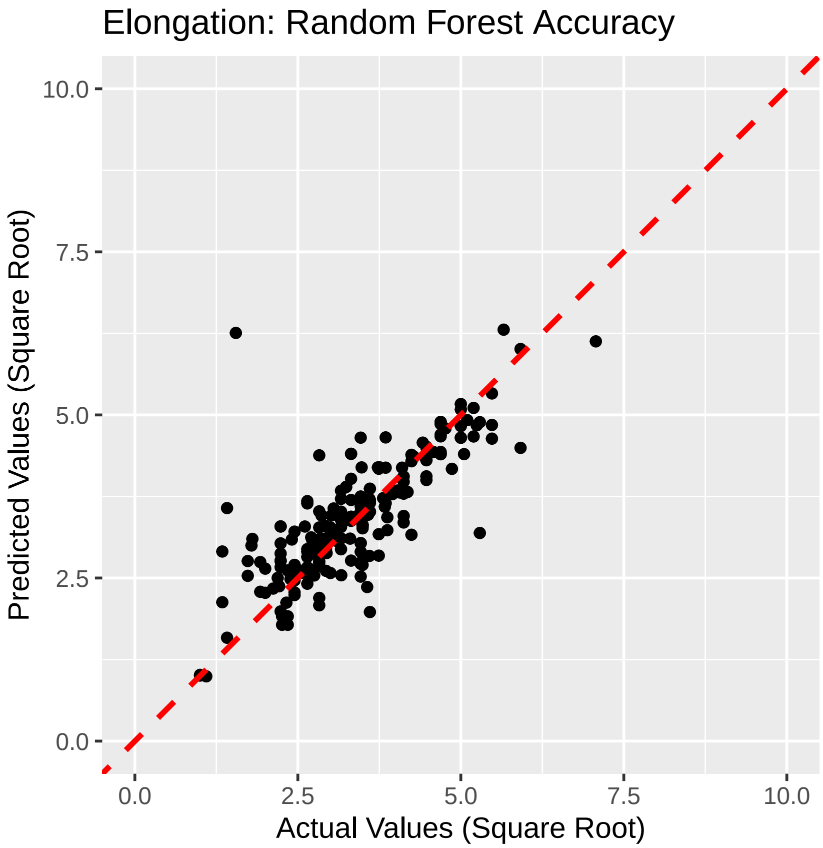
ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586

The next model is to try regression trees. After tuning the model provides a slightly worse training R^2 of 0.612 and a testing R^2 of 0.572.



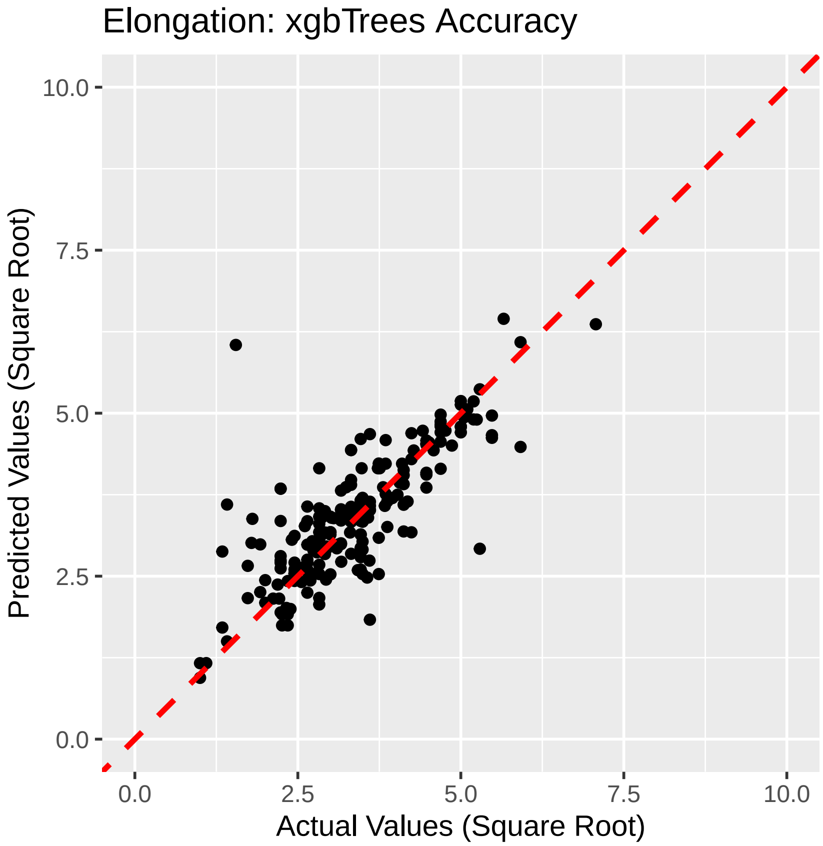
ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586
Regression Trees	0.612	0.572

The next model is to try random forest. After tuning the model performs much better providing a training R^2 of 0.740 and a testing R^2 of 0.634.



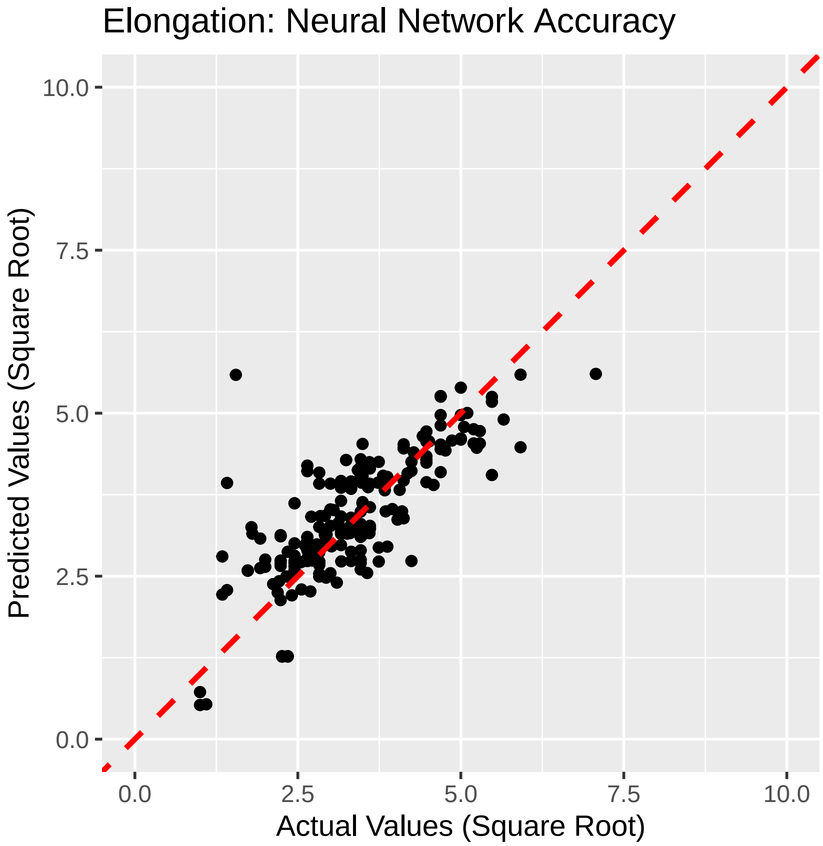
ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586
Regression Trees	0.612	0.572
Random Forest	0.740	0.634

The next model is to try extreme gradient boosted trees. After tuning the model also has relatively good results with a training R^2 of 0.691 and a testing R^2 of 0.626.



ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586
Regression Trees	0.612	0.572
Random Forest	0.740	0.634
eXtreme Gradient Boosted Trees	0.691	0.626

The next model is to try neural networks. After tuning the model provides a training R^2 of 0.607 and a testing R^2 of 0.603.

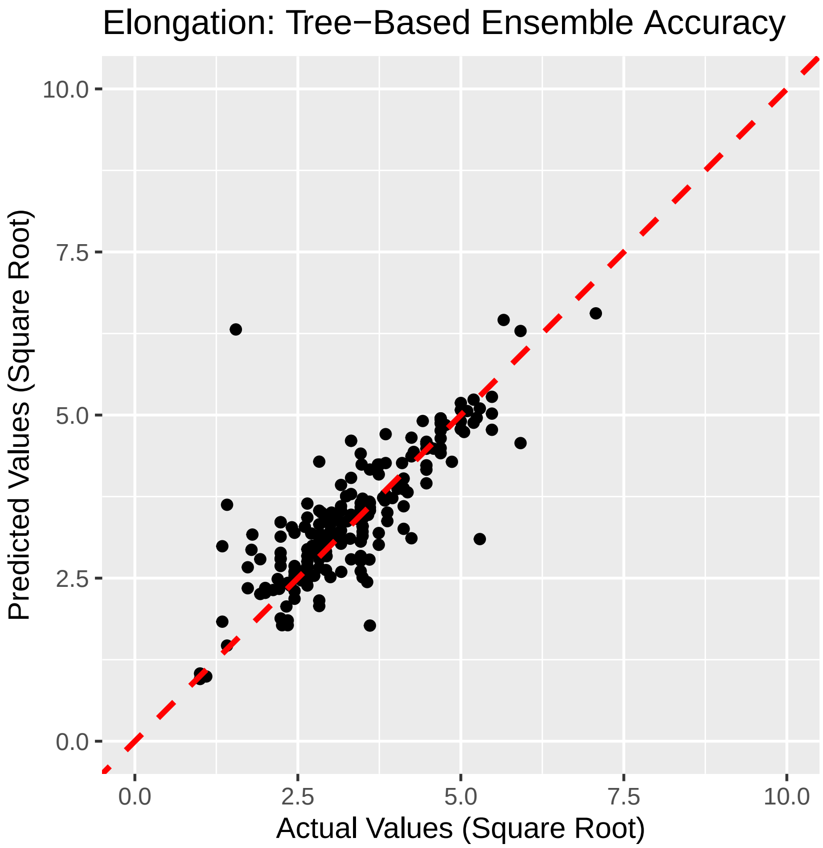


ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586
Regression Trees	0.612	0.572
Random Forest	0.740	0.634
eXtreme Gradient Boosted Trees	0.691	0.626
Neural Network	0.607	0.603

Some models on their own perform better than others, but combining some or all together into an ensemble model could produce better predictions than any one method alone. To evaluate the relative strengths of each method for predicting elongation, a generalized linear meta-model was applied to all six models and the resulting model importance summary was used to downselect the random forest and extreme gradient boosted trees tree-based models as the most dominant models that could be used for a final ensemble model.

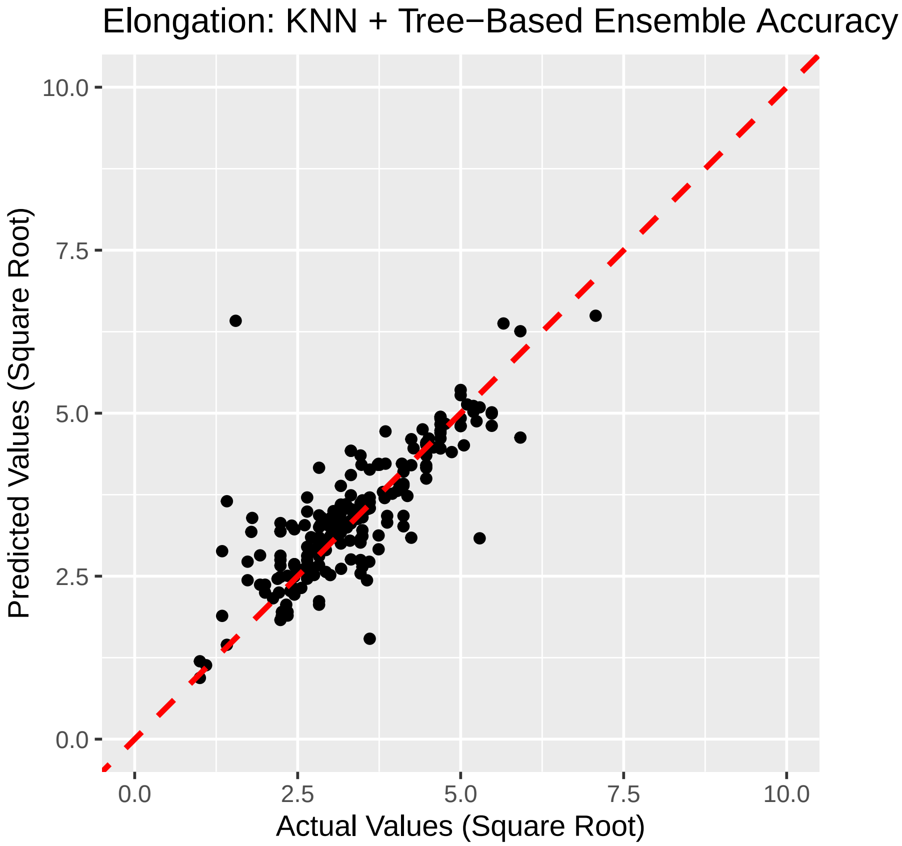
```
## The following models were ensembled: glm, knn, rpart, rf, xgbTree, nnet
##
## Model Importance:
##     glm     knn   rpart      rf xgbTree    nnet
##  0.0057  0.1259  0.0540  0.3833  0.4311  0.0000
##
## Model accuracy:
##    model_name metric     value         sd
##        <char> <char>     <num>      <num>
## 1:   ensemble   RMSE 0.4774289 0.06049415
## 2:        glm   RMSE 0.8407476 0.45156745
## 3:        knn   RMSE 0.6119346 0.05145368
## 4:      rpart   RMSE 0.7441491 0.05941504
## 5:         rf   RMSE 0.4958680 0.06823906
## 6:    xgbTree   RMSE 0.5121803 0.06383957
## 7:       nnet   RMSE 2.5401447 0.05353905
```

A downselected tree-based ensemble model provides the best training R^2 of 0.788 and a testing R^2 of 0.647.



ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586
Regression Trees	0.612	0.572
Random Forest	0.740	0.634
eXtreme Gradient Boosted Trees	0.691	0.626
Neural Network	0.607	0.603
Tree-Based Ensemble	0.788	0.647

A downselected tree-based ensemble model with KNN added into the ensemble also provides an almost equally good training R^2 of 0.790 and a testing R^2 of 0.640.



ElongationModel	TrainRsquared	TestRsquared
GLM	0.285	0.547
KNN	0.624	0.586
Regression Trees	0.612	0.572
Random Forest	0.740	0.634
eXtreme Gradient Boosted Trees	0.691	0.626
Neural Network	0.607	0.603
Tree-Based Ensemble	0.788	0.647
KNN + Tree-Based Ensemble	0.790	0.640

The tree-based ensemble is deemed the most appropriate model for predicting elongations due to its excellent performance in both training and testing. A more complex tree-based ensemble with KNN included is almost as good, but it may be best to restrict to just the tree-based components for model simplicity and less risk of overfitting. Random forest on its own is very close behind as an alternative non-ensemble model.

Results - Application of Models

Model development determined that tree-based ensemble models incorporating random forest and extreme gradient boosted trees algorithms were most appropriate and robust for the three mechanical properties of interest for aluminum alloys. Random forests are great for reducing variance in situations that could be sensitive to overfitting, while extreme gradient boosted trees are great for reducing bias from oversimplified relationships between variables [7]. Creating an ensemble between these methods helps incorporate the advantages of each method into one model that is more powerful than either method alone. To demonstrate application of the models, the original table can be filtered to just rows that contain at least one missing mechanical property and see how closely the model predicts the known values. Model outputs can be squared to get the final mechanical property predictions for comparison with known values using root mean square error (RMSE) as an evaluation metric. RMSE is calculated by taking the difference between predicted and actual ratings for each rating, squaring each residual, and taking the average of all the squared values. The square root of that value becomes a metric that represents the average error of the model. The following table summarizes how well the predictions match actual values in the incomplete rows alone:

Property	Rsquared	RMSE
Tensile Strength	0.971	24.049
Yield Strength	0.951	30.473
Elongation	0.931	2.297

Regarding R^2 performance all three models perform very well, with tensile strength performing best with $R^2 = 0.971$. This is followed by yield strength with $R^2 = 0.951$, and elongation with a strong $R^2 = 0.931$ despite being the worst performing of the three models. In comparing the RMSEs in context of the ranges of values covered in the provided data table:

- The range of tensile strengths in the provided table is 45-820 MPa, so an RMSE of 24.049 implies an approximate error of 3% of the total range covered by the dataset
- The range of yield strengths in the provided table is 10-790 MPa, so an RMSE of 30.473 implies an approximate error of 4% of the total range covered by the dataset
- The range of elongations in the provided table is 0.5-50%, so an RMSE of 2.297 implies an approximate error of 5% of the total range covered by the dataset

Even if not perfect, these errors are sufficiently low enough to provide well-informed and practical guidance on selecting alloys and manufacturing processes for a given engineering application. Given how well the models predict the known values, the models can then be used to fill in the unknown values with practical confidence.

The importance of each variable contributing to each model can be extracted from the tree-based ensemble models. The following table shows the top ten most important variables for each model that influence predictions:

Tensile Strength

Rank	Variable	Importance
1	Cu	100.000
2	Al	72.747
3	solutionisedTRUE	49.811
4	Mg	21.440
5	Zn	20.648
6	artificially_agedTRUE	15.330
7	Si	8.849
8	Zr	5.807
9	strain_hardenedTRUE	5.560
10	Li	5.336

Yield Strength

Rank	Variable	Importance
1	solutionisedTRUE	100.000
2	Cu	48.134
3	Al	43.427
4	artificially_agedTRUE	38.951
5	Zn	38.545
6	Mg	25.291
7	strain_hardenedTRUE	19.188
8	Zr	8.236
9	naturally_agedTRUE	7.969
10	series5	7.794

Elongation

Rank	Variable	Importance
1	Al	100.000
2	Si	79.478
3	series5	67.344
4	artificially_agedTRUE	66.669
5	solutionisedTRUE	54.443
6	Mg	46.594
7	strain_hardenedTRUE	40.131
8	naturally_agedTRUE	31.026
9	Cu	29.311
10	balance	27.473

As long as all elemental contents and binary process conditions are available as inputs, the models can be applied for property predictions. Considering that aluminum (Al) and copper (Cu) content are two of the most important predictors in the three models, perhaps a user of the models would like to see what happens if they adjust copper content by 1%. Start by taking an alloy of interest, add 0.01 to the Cu column, subtract 0.01 from the Al column, and make a joined table with the predictors. Apply the models to the table and square the results for comparison. The following example shows the difference in predictions with this composition change:

Alloy	AlPercent	CuPercent	TensileStrength	YieldStrength	Elongation
Original	0.8801	0.0198	634.3198	574.9856	16.5063
New	0.8701	0.0298	660.5313	595.4472	14.7079

Even a 1% change in the amount of two elements for an example alloy increases tensile strength by 26.2 MPa, increases yield strength by 20.5 MPa, and decreases elongation by 1.8%. This makes sense as copper is a known strengthening element for aluminum at the expense of some elongation and ductility [8].

Conclusion

The goal of this capstone project is to utilize a dataset of aluminum alloy elemental composition and processing condition combinations to predict tensile strength, yield strength, and elongation. For each of these mechanical properties, a tree-based ensemble machine learning model incorporating random forest and extreme gradient boosted tree methods was developed with effective results. Using these models, one can predict how an aluminum alloy's mechanical properties may change if slight modifications are made to the composition or if a different processing condition or temper is selected.

Alloy property exploration should be limited to small incremental elemental adjustments, as extreme changes may force the model to try extrapolating beyond what the model was trained on. For example, trying to adjust an alloy's composition from 1% copper to 2% copper is more likely to provide an accurate prediction of mechanical properties than trying to adjust from 1% to 25% considering that the maximum copper percentage found in the dataset is 7%.

The processing conditions take an oversimplified approach to temper classification, not closely taking into account process variables like heat treatment temperature, time left to age, and amount of strain hardening allowed. A more complete dataset with industry-recognized temper designations and process conditions would help provide the finer physical details that manifest into the decisions made in the models to generate predictions. Future work could involve refining the existing dataset with additional context for retraining.

Only six common machine learning algorithms were explored, but there are many more sophisticated algorithms left to explore that could result in better predictions. Optimization of each ensemble model's tuning parameters was also relatively minimal with opportunity to continuously improve performance with fine tuning and retraining with new alloys and measured properties. Future work could also be done on optimizing the tuning parameters in the models and further exploring other more complex tree-based methods.

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